

Newtrail Auto-Pilot: what has been tried in the last month, where we went in circles, and what to do now

Repository: **Zyugantax/Newtrail** (formerly himanshujena64-dot/Newtrail) | Feature: Auto-Pilot (topic to finished narrated video) | Period: 6 Jul 2026 (repo created) to 1 Oct 2026, with the focus on 15 Sep to 1 Oct | Prepared 1 Oct 2026

Sources read: all 422 pull requests, the commit history, the 80 CI workflows and 9,263 workflow runs, and every Auto-Pilot document in the repository: autopilot_collab (CURRENT_STATE, WORKLOG with 131 entries, DECISIONS D-001 to D-044, TEST_RUNS with 102 entries, MISTAKES M-01 to M-34 and P-01 to P-14, both Quality Ceiling plans, the Vercel audit), the PR #185 forensics (root cause report, 48-run failure study), the 11 percent crash analysis, the Claude/GPT handoff, the 1 Oct handover and the 127-point requirements document. No code was changed and no paid run was started for this report.

1. The short answers

- **What have we tried?** Everything that can be tried from the inside: 422 pull requests in 17 days, 1,801 hand-authored commits in September, a test suite that grew from about 1,300 to 2,280 tests, 44 recorded decisions, two multi-stage "quality ceiling" plans (13 + 20 items), a forensic study of 48 paid runs, a Director engine rewrite, a mistake ledger, a live progress channel, a cheap test ladder, and about 115 pipeline runs of which 80 bought paid media. Section 4 lists it all by date.
- **What did it produce?** Two complete videos. Both are the same 15-second, 2-scene Hindi Hanuman/Dronagiri short (25 Sep and 26 Sep). No other topic has ever finished. No part of the 5-minute Flydubai film has ever finished. The 48 paid runs of 22 to 25 Sep produced 0 videos. Of the 51 recorded run failures since 25 Sep, 19 were stopped by contradictions our own code wrote into the prompt and 17 by image QA.
- **Are we in a loop?** Yes, and the repository says so itself in several places: "the failure moved instead of disappearing" (root cause report), "removed the two exact delivered files' escape; it did not yet remove the class of escapes" (round 2 plan), "the regex blocklist approach is whack-a-mole" (worklog 30 Sep), and your own words recorded on 1 Oct: "same failure same money wasted" and "do not keep on going on loop". Twenty symptom classes recur across the month (section 5).
- **Were we misguided?** Partly, and in four specific ways: (1) the target was moved from a proven 2-scene short to a 45-scene real-incident film before the short was reliable; (2) attention was split, since roughly 60 percent of all pull requests were about the Vercel migration, landing pages, Stripe and UI, not Auto-Pilot; (3) every failure was answered by adding another AI gate or judge, although the 25 Sep study had already shown that stacked gates multiply failure (0.8 to the power 8 is 17 percent); (4) rules you set (no paid run before 9 of 10 plan-only passes, at most 2 paid attempts, \$38 cap) were written down and then bypassed within hours. Section 6 gives the evidence.
- **What to do?** Freeze scope, go back to the thing that works (2-scene shorts), and grow it in measured steps with a fixed pass-rate target and a fixed stop rule. Replace the tower of stochastic AI judges with fewer, calibrated checks and one human storyboard approval for long films. Spend money only on runs that are the final step of a \$0 ladder. Section 7 gives the ordered plan and the four decisions only you can make.

2. What Auto-Pilot is and what "done" means

Auto-Pilot takes a topic or brief and must deliver a finished, narrated video with no manual editing. The pipeline (18,571 lines in one file, autopilot_runner.py) runs: script writing (Gemini) to SceneTruth per scene (Gemini) to scene contracts to a Director plan (shot, lens, camera) to a final prompt gate (an LLM that reads every compiled prompt) to a cost preflight to images and START/END keyframes (Gemini image models, judged by two vision judges plus geometry) to narration (ElevenLabs) to video clips (Veo, judged again by majority vote) to stitch, export and a final QC. Images-only and plan-only modes stop early.

The goal you have stated, as recorded in the repository (D-032, D-033, the 127-point requirements): a 15-second film in about 15 minutes; plan-only batches passing 9 of 10 topics; living-being clip failures near 5 percent; first-time-right images rather than best of three; nothing hardcoded per topic; a fixed budget of about 5,000 rupees; and, since 30 Sep, one real 5-minute claymation Hindi film about Flydubai FZ1073 as the proof.

3. The month in numbers

Measure	Value	Where it comes from
Repository age	6 Jul 2026 (Streamlit template); first pull request 15 Sep	git log
Pull requests 15 Sep to 1 Oct	422 (about 100 in the last 5 days alone; 32 on 1 Oct)	GitHub
Pull requests about Auto-Pilot itself	roughly 140 to 160; the rest are Vercel migration (about 90), landing page, Stripe, UI, CI	PR titles
Hand-authored commits since 1 Sep	1,801 (plus about 2,700 bot progress commits)	git log
Busiest days	23 Sep: 231 commits; 25 Sep: 235 commits	git log
CI workflows / total workflow runs	80 / 9,263	GitHub Actions
Live Auto-Pilot smoke workflow runs	145	GitHub Actions
Long-film workflow runs	27 (one queued at the time of writing)	GitHub Actions
Pipeline runs recorded in TEST_RUNS	115 (67 with run ids since 25 Sep + 48 earlier paid runs)	TEST_RUNS.md
Runs that bought paid media	80 (32 since 25 Sep + 48 earlier)	TEST_RUNS.md
Complete videos delivered	2 (both 15 s, 2 scenes, same Hanuman topic, 25 and 26 Sep)	TEST_RUNS.md
Long-film parts finished	0 of 5 (furthest 70 percent; only one part ever reached video)	CURRENT_STATE.md
Stated settled spend since 25 Sep	\$38.72 (Flydubai film \$25.29 + cheap tests \$0.70 + measured runs \$12.7)	TEST_RUNS.md
Unpriced spend	48 earlier paid runs, all Gemini text and vision calls, ElevenLabs	TEST_RUNS.md, requirements doc
Owner budget	about 5,000 rupees (about \$56) total Google credit; \$38 media cap set 30 Sep	D-037
Decisions recorded / later amended, bypassed or reversed	44 / 12	DECISIONS.md
Mistake ledger entries in 48 hours (30 Sep to 1 Oct)	34 M entries + 14 P (process) entries	MISTAKES.md
Collaboration documents a new agent must read before touching code	about 800 KB across 9 files	autopilot_collab
Size of the main pipeline file	18,571 lines (12,625 on 25 Sep); changed in 251 commits since 20 Aug	wc, git log
Remote branches	330 (200 fix/, 30 audit/, 7 backup/ snapshots all taken on 20 Sep)	git branch -r
Long-film runs in the 18.5 hours from 30 Sep 21:27 to 1 Oct 15:58 UTC	25 (part 3 re-run in 15 of them)	autopilot-progress branch
Work redone on a moved main ("Reapply ... on latest main" commits)	about 20, on 21, 22 and 25 Sep	git log

Measure	Value	Where it comes from
Features re-added after the Vercel migration dropped them ("Restore ..." commits, 23 to 24 Sep)	about 20; the Veo frame-pair controls were restored three times	git log

4. Timeline: what was tried, phase by phase

Dates are 2026. PR numbers refer to Zyugantax/Newtrail. Run ids are GitHub Actions runs.

When	Phase	What was tried and what happened
6 Jul to 14 Sep	Build-up without pull requests	App created from a Streamlit template; direct commits only. Light activity until 26 Aug, then daily commits. Auto-Pilot already exists as a Streamlit wizard.
15 to 16 Sep	Stabilisation sprint and UI shell (PR 1 to 34)	P0 NameErrors in render/export, Auto-Pilot runtime reliability (R2), reliability sweep (R3), new Home and navigation shell, R2 multipart upload, YouTube bot-check fixes. Issue #29 "end-to-end stabilization checklist" opened 16 Sep, still open.
17 to 18 Sep	Audit fixes and buyer polish (PR 35 to 101)	A 50-bug end-to-end audit (audit/ branches); audit fixes #1 to #18; "second-pass" fixes; premium UI canvases; a UI merge reverted the same hour (PR 92 then 94); Veo shorts generation and stitch crash (PR 101).
19 to 20 Sep	Auto-Pilot crashes and the first "intelligence" layer (PR 102 to 148)	Crash fixes: QA schema KeyError, stitch NameError, re-entrant runs re-billing providers, final video not persisted, 0/100 scores. Then P0 to P5: fail-closed ceiling gates, SceneTruth Director, controlled START/END states, visual DNA continuity, temporal ledger, Supervising Director, Quality Learning Brain. Eight backup/ branches created on 20 Sep before risky fixes.
21 to 22 Sep	Progress-percent cliffs (PR 149 to 184)	Runs died at 4, 5, 6, 7 then 11 percent (the 11 percent crash analysis document). Fixes: grounding recovery, semantic QA truncation at 5 percent, FINAL_PROMPT_GATE 12-minute stalls, durable background jobs, resume after restart, guarded live 30 s smoke workflow (PR 181), one-shot forensic report (PR 180), pre-media gates collapsed into a bounded batch (PR 184).
22 to 25 Sep	48 paid runs, 0 videos; PR #185 held	All 48 benchmark runs used the solar eclipse topic. Funnel: 48 runs, 35 reached images, 7 reached Veo, 0 passed video QA. A forensic root-cause report said NO-GO and named the primary cause: truth is free prose re-serialised through about 12 string layers, so the first real check is a vision judge on paid pixels. GPT was brought in as an independent reviewer (handoff document, 25 Sep). Cost-aware acceptance bands designed (24 Sep).
22 to 28 Sep	Scope widens: Next.js/Vercel migration (PR 186 to 236, 276 to 314)	About 90 pull requests in parallel with the Auto-Pilot crisis: parity batches 01 to 29, certifier, Stripe checkout, Hook Lab, Help pages, Web Editor; five landing-page redesigns (PR 287, 315 to 319, 322, 324, 326 to 328, 348).
25 Sep	First full success	PR #185 merged after 3 days; PR 240 to 243 (self-healing, Pandas Series crash). Three live runs each hit a different blocker (eclipse image QA twice, then a false-positive gate). Veo overshoot fix (PR 261, 263) and the gate fix (PR 265). Run 36153764328: 15 s Hanuman short delivered end to end, about 21 minutes.
26 Sep	Second success, faster	Latency work (PR 312): stage timing, concurrent judges, 2 Veo scenes in parallel, targeted QC repair. Run 36266614262: same Hanuman topic in 11.7 minutes, 4K vertical. This is the last complete video to date.
27 Sep	Topic robustness: 5 of 5 fail	Gandhi (Veo safety filter), honeybee (END keyframe 3 of 3), roti (START drawn finished 3 of 3; start-habit-review truncated), kingfisher twice (gate contradiction; START 3 of 3). Fixes collected in draft PR 323. Devanagari plan-name fix (PR 321).
28 Sep	Vercel Auto-Pilot audit and Quality Ceiling plan 1	Audit: production ran code known to fail; a pre-spend gate was dead in all five runs; a Veo billing trap; two security holes; money spent on calls that could not work. A "judged design panel" produced 13 changes. First batch (1 to 6, 8, 10) merged as PR 349 the same day. CI moved to a self-hosted Windows runner (PR 347).
29 Sep	Quality Ceiling round 2: 20 more items in 7 stages	An adversarial audit of the day-old PR 349 found 38 confirmed defects in it and the surrounding code. Stages 1 to 7 merged as PR 350 to 361 within 36 hours. The one approved paid run (36579085454, \$2.32) was rejected by the new typed judges (fused anatomy). D-025 granted standing approval for 3 paid runs of \$8; never completed.
30 Sep (day)	Director v2, Gemini truncation root cause, batch loop	Director engine rewritten (D-026). Gemini thinking tokens were eating the output budget at 36 call sites (D-027). Plan-only batches of 10 topics: 7/10, then 8/10, then 8/10 (target 9/10 never reached). CI moved again, to Blacksmith runners (PR 362 to 371). Measured paid run 36672474437 (\$4.27) rejected twice on wrong endpoint; 36732743162 (\$5.87) rejected 3 times.

When	Phase	What was tried and what happened
30 Sep (evening)	The long film begins	You approved the Flydubai FZ1073 film: 5 minutes, 45 scenes, claymation, Hindi, real people shown faceless, \$45 then \$38 media cap, max 2 paid attempts (D-037). The 48-scene plan could not fit one Gemini call (M-04, M-06), so the film was split into 5 parts of 9 scenes. Mistake ledger created (D-036). Paid parts 36777268715 and 36782036148 failed on quota starvation, hand-count scenes and camera text in truth.
30 Sep night to 1 Oct 04:00	Twelve more part runs, one scene at a time	M-08 to M-25: camera direction (three separate fixes), identical START/END, narration too long, faceless rule on an empty podium, continuity ledger, re-plan contradictions (five times in one night), judge not repeatable (M-24). \$25.29 sunk on the film with no finished part. 03:55 UTC you stopped paid runs: "I do not have that much money for wastage".
1 Oct	Cheap ladder, owner review, prompt adherence	Cheap image mode (D-039). You reviewed 24 keyframes: 13 OK, 11 redo, 9 of them wrong uniform or invented person; the automatic judge had passed all 24 (M-26). Root cause M-33: the image model never received the author's scene text, only machine contract markup. Script lint before dispatch (D-041). CAST clause fixed twice (M-29, M-34). AP-001 to AP-007 hardening. Handover and 127-point requirements documents written. Plan-only: 4 of 5 parts pass text gates. An images-only run of the whole film is queued.

5. The loop, with evidence

A loop is the same class of problem returning under a new name after a fix. The worklog, mistake ledger and test register show twenty such classes. The most important are below, with the dates on which each was "fixed".

Recurring class	When it was fixed	Times
Our own compiler writes a contradiction that our own gate then refuses	25 Sep (#92, #99, #103); 30 Sep four times (static clause, "whack-a-mole", allowlist D-029, batch round 1 M-01); 1 Oct twice (M-33 markup, M-34 CAST clause)	7+
Camera direction leaking into SceneTruth text	30 Sep M-08 (placement), 30 Sep M-09 (same night, direction), 1 Oct M-18 (serializer still printed it)	3
Gemini reply cut off because thinking tokens use the output budget	25 Sep (#94); 27 Sep roti and kingfisher ("budget raised again", "fitting was luck"); 28 Sep audit (dead in 5 of 5); 30 Sep D-027 at 36 sites; 30 Sep M-04 script writer; 30 Sep M-06 SceneTruth ("third time")	6
Image model draws the END state as START (drawing habit)	25 Sep (#86, #89, #100, #105 "again"); 26 Sep (two runs, run #115); 27 Sep (roti, kingfisher, D-017 re-plan)	6+
Judge false pass or false fail; verdict not repeatable	25 Sep; 26 Sep (tie-break flags); 28 Sep (6 s frozen frame passed two judges); 29 Sep (R2-2, R2-4, R2-14); 30 Sep (M-03); 1 Oct (M-24 accepted image rejected on re-run; M-26 judge passed 10 of 24 you rejected)	7
Keyframe re-plan creates a new contradiction instead of fixing the old one	30 Sep M-13 (twice in one night), 1 Oct M-20, M-23, M-24 ("fifth time tonight"); 0 successes in 5 attempts	5
Cast, wardrobe, identity, anatomy mismatch	28 Sep (changes 5, 6, 8); 29 Sep (typed morphology, Stage 4, Stage 5); 30 Sep; 1 Oct (M-26, M-29, M-31, M-32, M-34, AP-036): the cast check was introduced on 1 Oct and revised four times the same day (PR 407, 409, 411, 421, 422)	9
Narration longer than picture (freeze, clone, cut)	28 Sep (changes 1, 10); 29 Sep (R2-1, R2-3: cap was a no-op in Hindi); 30 Sep (D-024 withdrawn 4/6 s clips, D-030); 1 Oct (M-15)	5
Image quota 429 starvation	25 Sep; 26 Sep; 30 Sep three times (M-10 patience, M-12 max_parallel, recurred because parts were dispatched in separate runs, P-13)	5
Cost preflight wrong (prices fiction, cap not enforced)	28 Sep audit; 29 Sep R2-16; 30 Sep (table below Vertex sheet; duration always 8 s); 1 Oct M-30 then PR 410 the same day; AP-002, AP-003	6
Judge outage buys media again	24 Sep forensics R3; 25 Sep claimed fixed on PR #185 branch (never merged); 28 Sep audit A3/A8/A9 still present; 29 Sep R2-14 still open	4
Retry re-serves the same prompt or the same rejected bytes	27 Sep (3 of 3 identical failures); 28 Sep audit A2 ("fresh" resample is a cache hit); 29 Sep R2-15 tombstones; 30 Sep "the retry changed only the wording"	4
Live run blocked by a new, unrelated pre-spend blocker	25 Sep ("three consecutive attempts, each a different blocker"); 29 to 30 Sep (NameError, 8 s limit, quota, wording twice, wrong endpoint, scene length, quota again); 30 Sep to 1 Oct (12 part runs, each a new scene)	continuous
CI runner churn	ubuntu to self-hosted Windows (28 Sep, nine follow-up fixes, 3 to 4 hour CI rounds, dropped jobs) to Blacksmith (30 Sep, 74 workflows, five PRs)	2 migrations in 3 days

Rules written down and then not followed

- **D-033 (30 Sep):** no paid measured run until a plan-only batch passes 9 of 10. Batches reached 7, 8 and 8 of 10. Paid long-film runs started the same evening under D-037.
- **D-037 (30 Sep):** max 2 paid attempts on the film. Parts were re-dispatched at least nine times before you stopped it.
- **D-002 / D-016:** no paid run without a specific approval and an offline review of the failing stage first. D-025 then granted standing approval for three runs; several more followed under the film decision.
- **D-036 (30 Sep):** "never ship the same kind of patch a third time". M-18 was the third camera-direction patch; M-34 was the second CAST-clause patch within hours; the keyframe re-plan failed five times in one night.
- **Handoff (25 Sep):** "Do not make broad architecture changes unless failure evidence points to architecture as the cause." Three days later a 13-change plan and a 20-item round 2 were merged in 36 hours, followed by a Director rewrite.

6. Have we been misguided? An honest assessment

The fixes were real. The test suite went from about 1,300 to 2,280 passing tests, true bugs were found (a `NameError` that cost a paid Pro call on every run, a cache that returned the rejected image on every "fresh" retry, a judge that bought a new clip when it was merely unavailable, a prompt that never contained the author's text). Nothing in the record suggests laziness or bad faith. The misguidance is in direction and method, in four ways.

6.1 *The target moved before the base was stable*

- The pipeline has succeeded exactly twice, on 2 scenes of one simple mythology topic. On 27 Sep five ordinary topics failed. The repository's own forensics state that every batched stage "was designed and proven at about 8 scenes or fewer".
- On 30 Sep the target became a 45-scene, 5-minute, real-incident film with two real people who must be shown faceless, in a style (claymation) never delivered before, in Hindi, with hand-to-hand struggle scenes that the image model cannot draw (M-13). That is a jump of roughly 20 times in scene count and several steps in difficulty, taken on the day the 10-topic plan-only batch was still at 8 of 10.
- Result: 12 part runs, \$25.29, 0 finished parts, 27 mistake-ledger entries in 48 hours, and most of the fixes were hand edits to the one script (requirements doc point 117), which do not carry to the next topic.

6.2 *Attention was split across a product, not a feature*

- Between 22 and 28 Sep, while Auto-Pilot had 0 of 48 videos, roughly 90 pull requests built a Next.js/Vercel copy of the whole Streamlit app (parity batches 01 to 29, a certifier, Stripe, Hook Lab, Web Editor), and five landing-page redesigns were merged in two days.
- The Vercel path then needed its own Auto-Pilot audit (28 Sep): a Veo billing trap, two security holes, a download button that always returned 404. The migration created a second place where Auto-Pilot could fail.
- Of 422 pull requests, fewer than 160 are about Auto-Pilot. Every one of the others consumed review time, CI time and the attention of the same person.

6.3 *Every failure was answered with another AI gate*

- The 25 Sep failure study computed the problem exactly: with eight strict gates at 80 percent each, the end-to-end pass chance is 17 percent; each gate would need 97 percent for a usable rate. It listed "too many compounding strict gates" as root cause 2.
- The response over the next five days added gates: typed hard defect classes, a pair continuity judge, a cast check, a focused wardrobe check, a final-gate repair loop, majority-vote judges, a \$0 motion gate, a continuity ledger, a start-habit review, a master-script lint. Several of these are themselves LLM calls with their own truncation and variance problems (M-03, M-24, D-027).
- The judges are not repeatable (M-24) and disagree with you (M-26: the pipeline had accepted all 24 saved images; you rejected 11, ten of them for the wrong person or uniform). A gate that is wrong a third of the time, stacked eight deep, is the "reject-and-recreate loop" you described. Adding a ninth gate cannot fix that; removing or calibrating gates can.
- About 20 to 30 Gemini text and vision calls happen before the first Veo call. That is why 429 quota errors are "constant in every run" (forensics) and why planning alone takes 10 to 20 minutes per attempt, so each lesson cost a 20-minute run (P-14).

6.4 *Process weight grew faster than results*

- A new agent must read about 800 KB of collaboration notes before touching code, and update five of them after every change. Four pull requests on 1 Oct contain nothing but a "morning summary". The NEXT ACTION pointer was superseded nine times in 36 hours.
- The main file grew from 12,625 to 18,571 lines in one week while the problem list did not shrink. Fixes made on branches that were never merged (the PR #185 side branch) were re-found as new defects on main three days later.

- Two CI runner migrations in three days cost nine follow-up fixes, 3 to 4 hour CI rounds and dropped jobs.
- Parallel sessions overwrote each other: about 20 commits are titled "Reapply ... on latest main", seven are exact duplicates of an earlier commit, and the same upload fix exists as ten branch variants (fix/hf-space-upload-403 with suffixes -one, -real, -work, -tdd, -final). Work was being done twice because main moved underneath it.
- Decisions were recorded faithfully and then overridden by the next evening's urgency. A rule that is bypassed the same day it is written is not protecting you.

Verdict: the loop is not caused by one bad engineer or one bad model. It is caused by chasing a moving, oversized target with a method (add a gate, re-run, read the new failure) whose cost per lesson is 20 minutes and a few dollars, and whose gates are themselves unreliable. The system has proven it can make a 15-second short. It has not proven anything else, and the current approach does not converge towards proving it.

7. What exactly to do now

The order matters. Steps 0 to 3 cost no paid media. Do not start step 4 until step 3 is green. Do not touch the long film until step 6.

Owner instruction, 1 Oct 2026: **no money is to be spent on video for now.** Steps 4 and 5 below are therefore parked until you lift that instruction in writing; steps 0 to 3 and 7 need no video spend and are the work to do now. Note: at the time of writing a long-film images-only run (36887831784) that you dispatched at 15:53 UTC is queued; it buys images, not video, and is yours to cancel or keep.

#	Step	Detail
0	Decide the goal in one sentence and freeze scope for two weeks.	Recommended: "Auto-Pilot delivers a 15 to 30 second short on any ordinary topic, first time, in under 15 minutes, for under \$4, at least 8 times out of 10." Everything else (Vercel, landing pages, Stripe, long film, Kling, HF Omni, Director Studio) is frozen until that sentence is true and measured. Archive or leave the Vercel branch as is; it does not need to be deleted, only left alone.
1	Cut the judge tower down to what you can calibrate.	Run judge-calibration.yml against your 24 labelled keyframes today (it exists, costs cents, has never been run). Keep a check only if it agrees with you on at least 80 percent and wrongly rejects at most 2 OKs (your own D-042 bar). Remove or make advisory every check that fails this bar. Target: at most 3 gates before an image is bought (deterministic lint, one contract check, one image judge) and 2 after a clip is bought (motion metrics, one semantic judge). Record the before/after gate count in CURRENT_STATE.
2	Build a \$0 regression corpus from the artifacts you already paid for.	You have 13 Veo clips, about 100 keyframes with verdicts, 24 owner labels, and saved prompts from runs 36820254640 and 36826423299. Turn them into fixtures (this was begun in tests/test_still_prompt_markup.py). Every judge and compiler change must replay against this corpus offline before any live run. This is how a lesson stops costing 20 minutes and money.
3	Measure topic generality with plan-only batches and hold the 9 of 10 bar.	Ten ordinary topics (not eclipse, not real people), plan-only, until 9 of 10 pass the text gates. The batch workflow exists. If two rounds stay below 9 of 10, D-033 already says what to do: simplify the pipeline, do not patch the checks. Keep that promise this time.
4	One measured paid run at a time, on the proven 2-scene shape, three different topics.	Cap \$5 each. Read the artifact fully before the next dispatch. Success means a QC-passed MP4 you have watched. After three successes on three topics you have, for the first time, evidence beyond Hanuman. After any failure, write the M entry and fix the class offline against the step-2 corpus before spending again.
5	Grow scene count in steps: 2, then 4, then 8.	Each step repeats step 4 (three topics, pass rate recorded). Stop growing the moment the pass rate falls below 8 of 10. This gives you a measured ceiling for how long a film the system can make unattended.
6	Long films only with a human storyboard approval step.	A 45-scene real-incident film is beyond what an unattended pipeline has shown it can do. Make it a two-pass product: the system writes the script and plans every scene, you approve the storyboard and cast sheet once (as you did with the 24 keyframes), then the system renders parts with the approved keyframes locked (D-040). This removes the single largest source of loops (re-plans and judge variance) from the expensive stage.
7	Restructure the work so lessons stay general.	Split autopilot_runner.py into stages (planning, compile, image, video, assembly) with one test file each. Replace per-film hand edits with script-writer rules (requirements points 118, 127). Keep one page of CURRENT_STATE and let the rest be history. Stop writing summary-only pull requests.

Stop rules (write these into DECISIONS and keep them)

- No paid run unless it is the last rung of a ladder that passed: offline corpus, then plan-only, then images-only, then video. Your own D-039 ladder, applied every time.
- A class that recurs twice is fixed by simplifying or removing, never by a third patch (D-036). If the third fix is being written, stop and change the approach instead.
- No pull request that mixes Auto-Pilot engine changes with UI, Vercel, CI or documentation changes.
- One live run at a time, read fully before the next. No overnight standing approvals.
- If after two weeks the step-4 pass rate on three topics is below 8 of 10, the architecture is the problem (prose truth re-serialised through 12 layers, as the 24 Sep forensics said). Then the next step is the typed ScenePlan and single audited request from that report, built on a branch and replayed against the corpus, not more gates.

Decisions only you can make

- Is the goal reliable shorts, or the 5-minute film? Section 7 assumes shorts first and the film as a supervised product. If the film is the only goal, say so, and step 6 becomes step 1 with a human approval stage built in from the start.
- Which checks may be removed or made advisory when calibration shows they disagree with you? (D-003 forbids lowering thresholds without your word. Removing an unreliable judge is a different decision and needs your word too.)
- How much of the remaining roughly 2,800 rupees is for measurement (steps 4 and 5) and how much is reserved for the film?
- Is the Vercel migration paused until the shorts target is met?

8. Your own words, as recorded in the repository

These are quoted from WORKLOG.md, MISTAKES.md and CURRENT_STATE.md. They are the clearest statement of the goal and of the loop, and they should be the first page any future session reads.

"we shall not again go into a loop; always re-think what mistake happened and how to correct it to reach the goal." (30 Sep, became D-036)

"rather than waiting every time 45 min, 5 hours" (30 Sep, became D-035 live progress)

"I do not have that much money for wastage .. find some cheap solution for testing" (1 Oct, 03:55 UTC; paid runs stopped)

"these are so silly mistake ... it feels like we are just manipulating not learning from our mistakes" (1 Oct, became P-14 and the script lint)

"we have to do more prompt adherence .. not every time best of 3" (1 Oct, M-29)

"do not be on loop that rejecting and creating" (1 Oct)

"nothing need to be hardcoded because video topic will be keep on changing" (1 Oct, requirements point 20)

"same failure same money wasted" (1 Oct, worklog M-33 entry: "21 days")

"majorly true, not very strict" (on planning strictness, requirements point 94)

9. What is good and should be kept

- The two successful runs and their exact code: PR 265 (gate false positive), PR 261/263 (end-state hold), PR 312 (latency). This is the proven core.
- The mistake ledger discipline (evidence, class, why it got through, correction, recurrence, measured effect) is the right instrument. It only needs to be obeyed on the recurrence rule.
- The cheap ladder (plan-only, images-cheap, short video smoke, full) and the live progress channel.
- Owner-approved keyframes locked and never re-judged (D-040). This is the seed of the human storyboard approval in step 6.
- The master-script lint that runs before any part at \$0 (D-041), as long as every new class gets a rule in the same commit.
- The forensic documents. The 24 Sep root-cause report already contains the architecture the plan in section 7 falls back to if gates alone cannot reach the target.

Appendix: the two recorded successes

Run	Date	Topic and shape	Result
36153764328	25 Sep	Hanuman lifts Dronagiri; 15 s; 2 scenes; Hindi; Veo; cap \$3	14.8 s, 2160x3840 MP4, QC passed, about 21 min, no retries
36266614262	26 Sep	Same topic and shape; after latency PR 312; cap \$5	15.1 s, 2160x3840 MP4, QC passed, 11.7 min, both Veo scenes concurrent